

THE INSIGHT ARCHITECTS

Predicting Contraceptive Use in Kenya

What 32,156 women in the 2022 Kenya Demographic & Health Survey tell us about who uses family planning — and who is being left behind.

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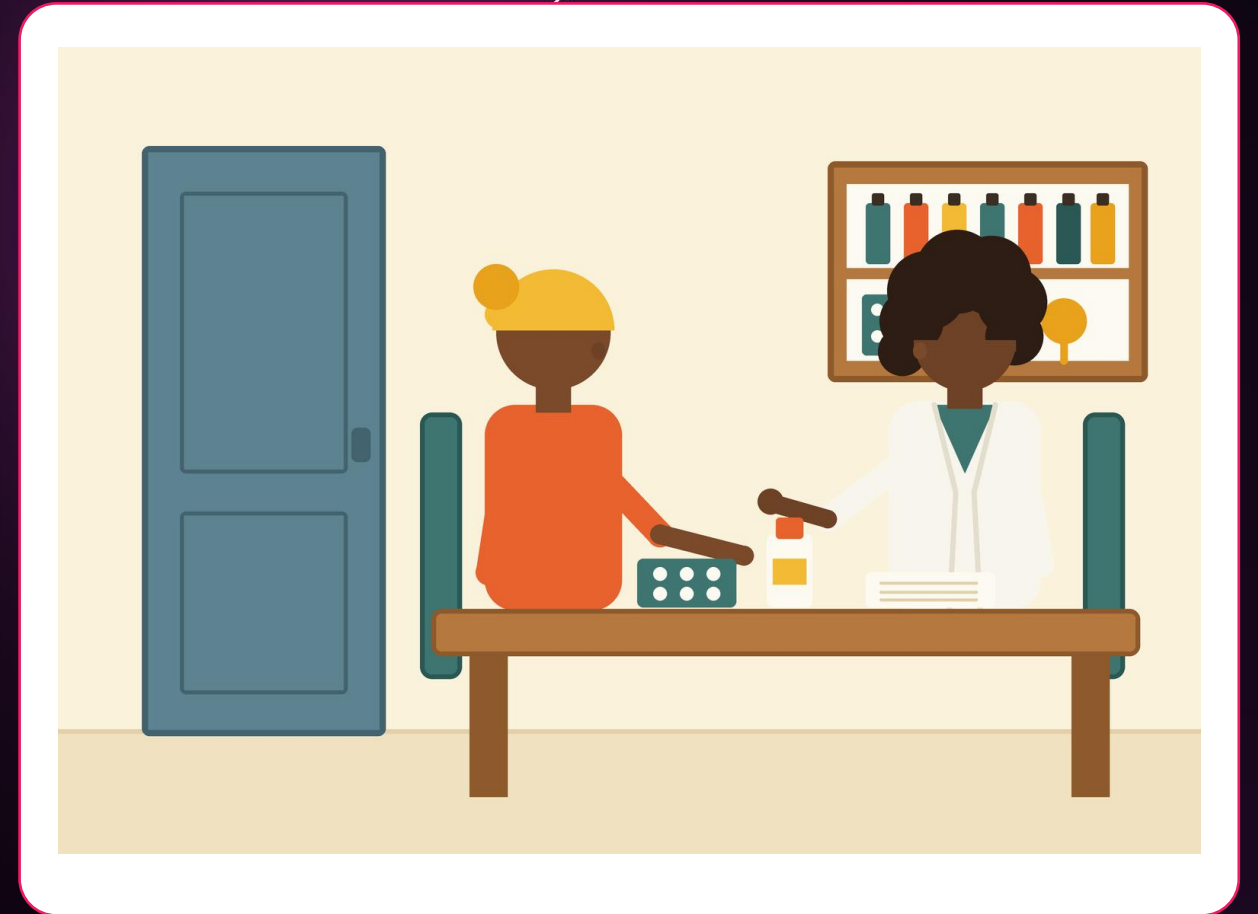
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Live tool: kdhs-contraceptive-api.onrender.com



WHY THIS PROJECT EXISTS

One country. Two realities.

57%

EMBU COUNTY

of women aged 15–49 use modern contraception

VS

1.2%

MANDERA COUNTY

of women aged 15–49 use modern contraception

WHY IT MATTERS

Behind every data point is a real woman navigating her choices within the realities of her county and household. **A 56-point gap between two counties in one country means help is not reaching the women who need it most.** This project predicts who is being left behind, explains why, and puts that knowledge into planners' hands.

OUR OBJECTIVES

Five questions we promised to answer

Our mission, in plain words: help Kenya's health planners find the women family planning is not reaching — and give them a working tool to act on it.

Q1

THE PICTURE

How many women use a modern method, a traditional one, a folkloric one — or none at all?

Answered in Part 2

Q2

PLACES & PEOPLE

Which counties have the most non-use, and what do the women at highest risk have in common?

Answered in Parts 2 & 5

Q3

MONEY vs SCHOOL

How do wealth and education work together to shape whether a woman uses contraception?

Answered in Part 2

Q4

TOWN vs COUNTRY

Does living in a city or the countryside still matter once wealth and education are taken into account?

Answered in Part 2

Q5

THE PREDICTOR

Which prediction model works best — and what actually drives its decisions?

Answered in Parts 3 & 4

Watch for the pink tags on the slides ahead — each one marks a question being answered.

How we'll walk you through it



PART 1

THE DATA

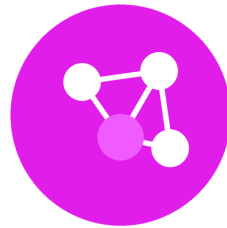
One national survey, 32,156 voices



PART 2

WHAT IT SHOWS

The patterns that answer Q1–Q4



PART 3

THE MODEL

Five approaches compete on fair ground



PART 4

WHY IT DECIDES

Every prediction, explained



PART 5

ACTION

Answers become fundable programmes

Method: CRISP-DM · Tools: Python · scikit-learn · LightGBM · SHAP · Flask

PART 1 OF 5

The Data

One survey knocked on doors in all 47 counties and asked women about their lives. Everything in this project starts with their answers.



32,156 women, in their own words

32,156

women interviewed face-to-face

47

counties — every one in Kenya

15–49

age range of respondents

5,925

raw survey variables, distilled to the ones that matter

WHAT'S HERE

The 2022 Kenya Demographic & Health Survey (KDHS) — women's questionnaire. For each woman: age, county, education, household wealth, religion, work status, marriage, children born and living — and which contraceptive method, if any, she currently uses.

62% of respondents live in rural areas, 38% in urban ones, and every wealth group holds roughly a fifth of the sample — so every kind of woman is fairly represented.

WHY IT MATTERS

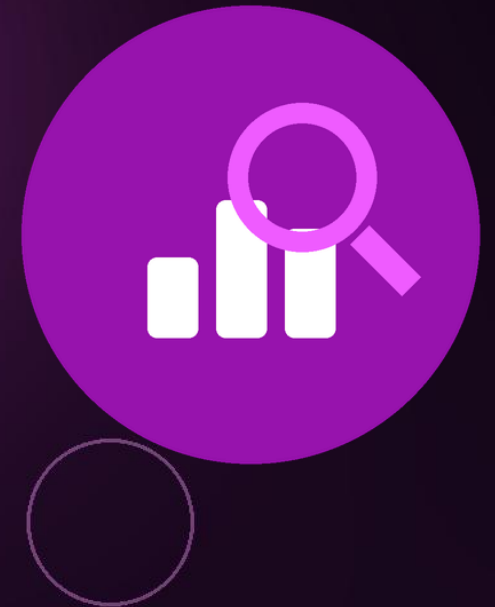
It is the most detailed, nationally representative snapshot of Kenyan women's reproductive health available — the same lens the Ministry of Health uses to plan. A pattern found here describes the real Kenya of 2022.

Honesty notes: answers are self-reported, only women were interviewed, and one snapshot in time shows associations, not proof of cause.

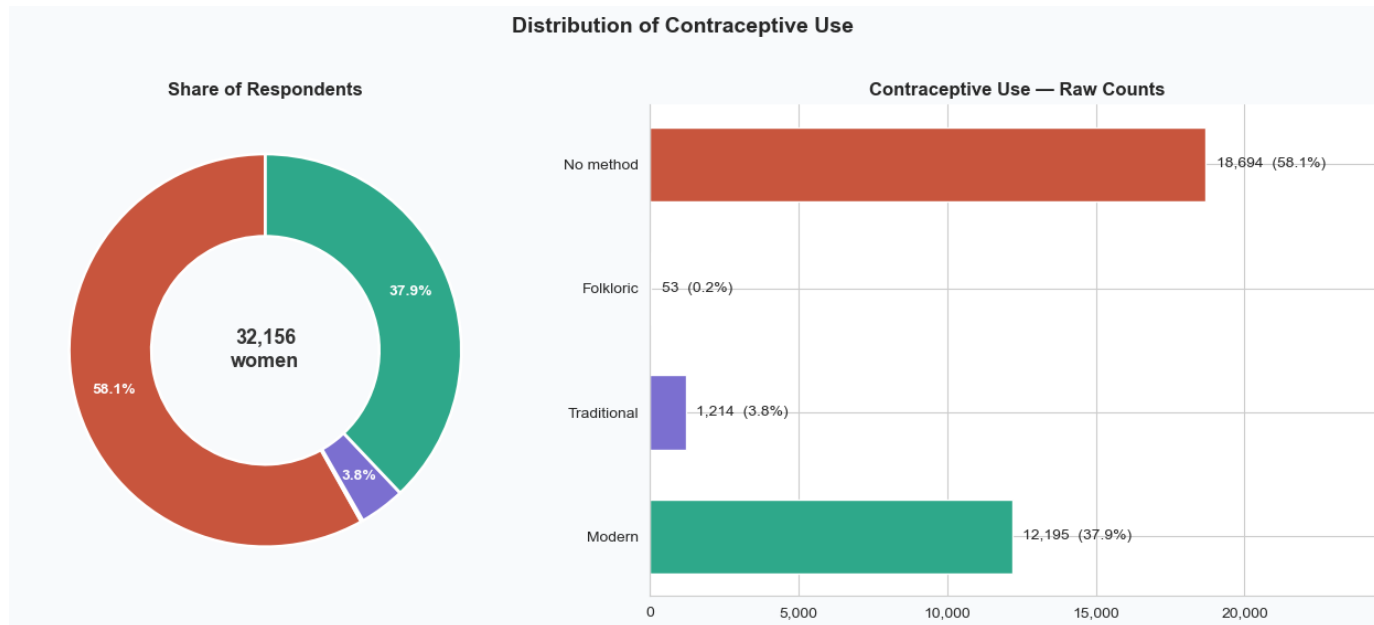
PART 2 OF 5

What the Data Shows

Before any model, we let the data speak — and it answers four of our five questions directly.



Who uses what, today



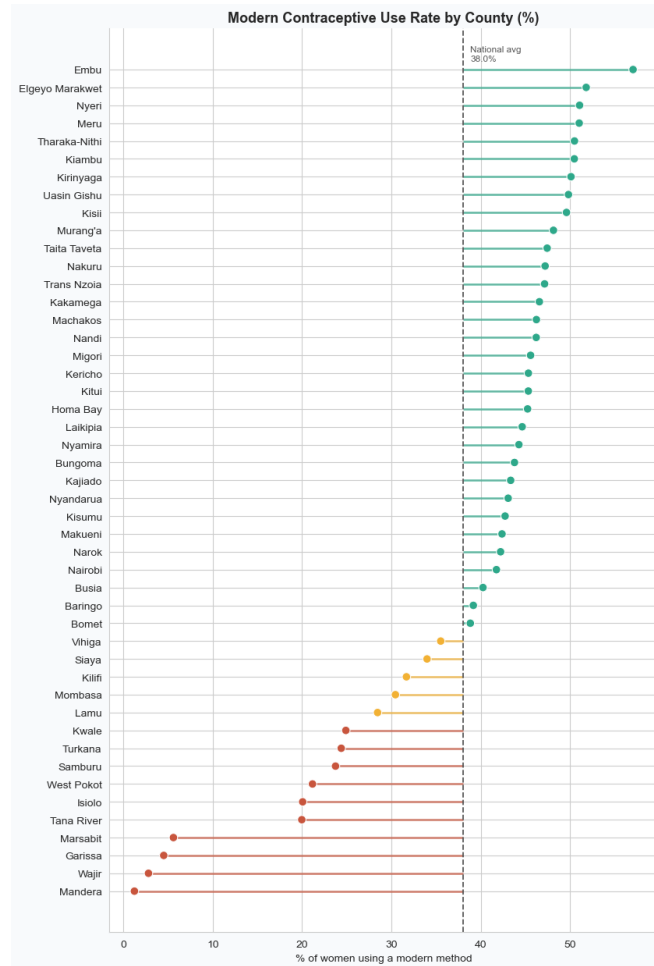
WHAT YOU'RE SEEING

58.1% of women use no method at all. 37.9% use a modern method. Traditional methods cover 3.8%, and folkloric methods just 53 women — 0.2% of the whole survey.

WHY IT MATTERS

This answers Q1 — and it sets the rules for everything after. Two groups dominate; two are vanishingly rare. A model can learn a lot about the big groups, but no method on earth can learn reliably from 53 examples. We balance the training data and judge models with metrics that respect the small groups.

Geography is the biggest divide



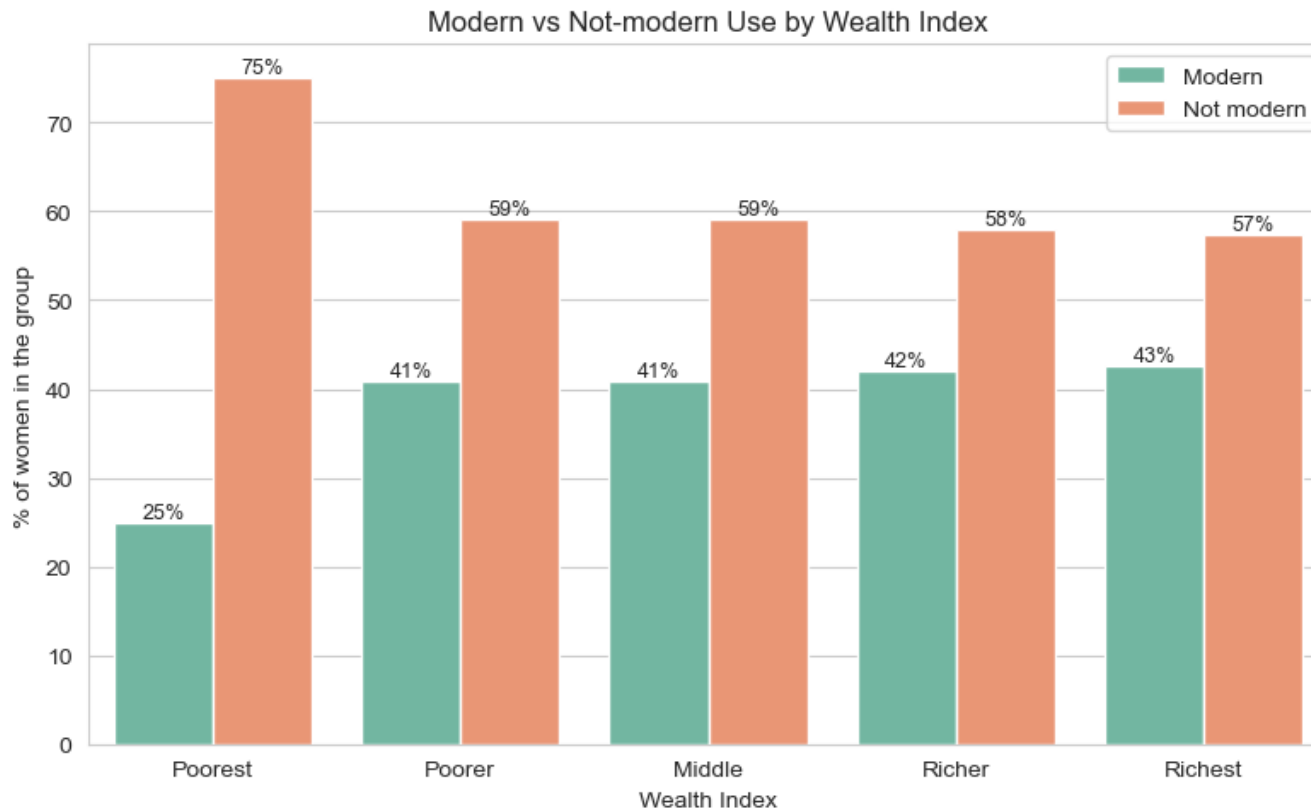
WHAT YOU'RE SEEING

All 47 counties, ranked by modern-method use. Embu leads at 57%; Mandera sits at about 1%, with Wajir, Garissa and Marsabit all under 6% — the whole arid north clusters at the bottom, far from the 38% national average.

WHY IT MATTERS

This is the heart of Q2. The counties with the greatest need are knowable in advance — so family planning can be targeted by place. A uniform national programme treats Embu and Mandera the same; this chart shows those two situations have nothing in common.

Wealth is a cliff, not a ladder



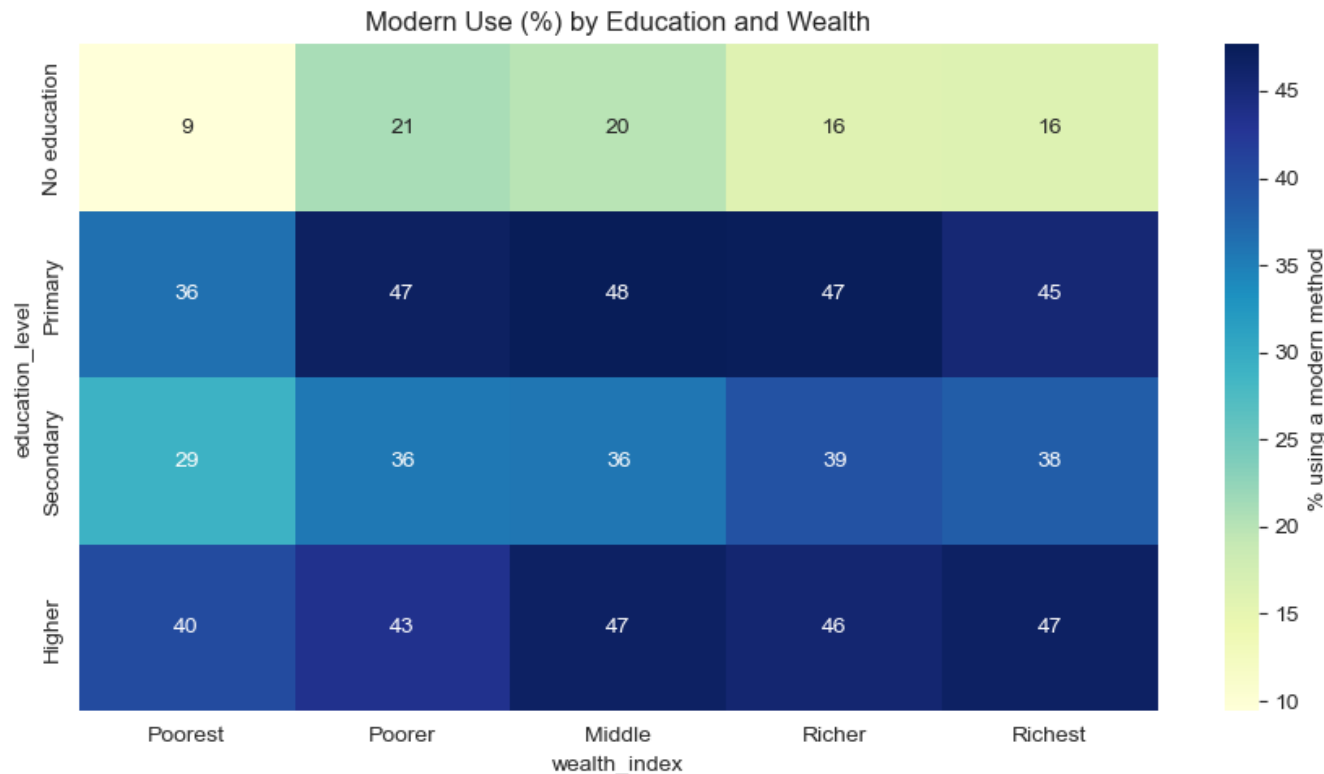
WHAT YOU'RE SEEING

The poorest women stand alone at 25% modern use. Every group above them is nearly flat — 41%, 41%, 42%, 43%. The big gap is between the poorest and everyone else, not a steady climb from poor to rich.

WHY IT MATTERS

If money mattered evenly, the bars would climb like a staircase. They don't. **Almost the entire wealth effect happens at the very bottom** — so subsidies and free supplies aimed at the poorest fifth of households buy far more impact per shilling than spreading the same money across everyone.

Education outweighs wealth



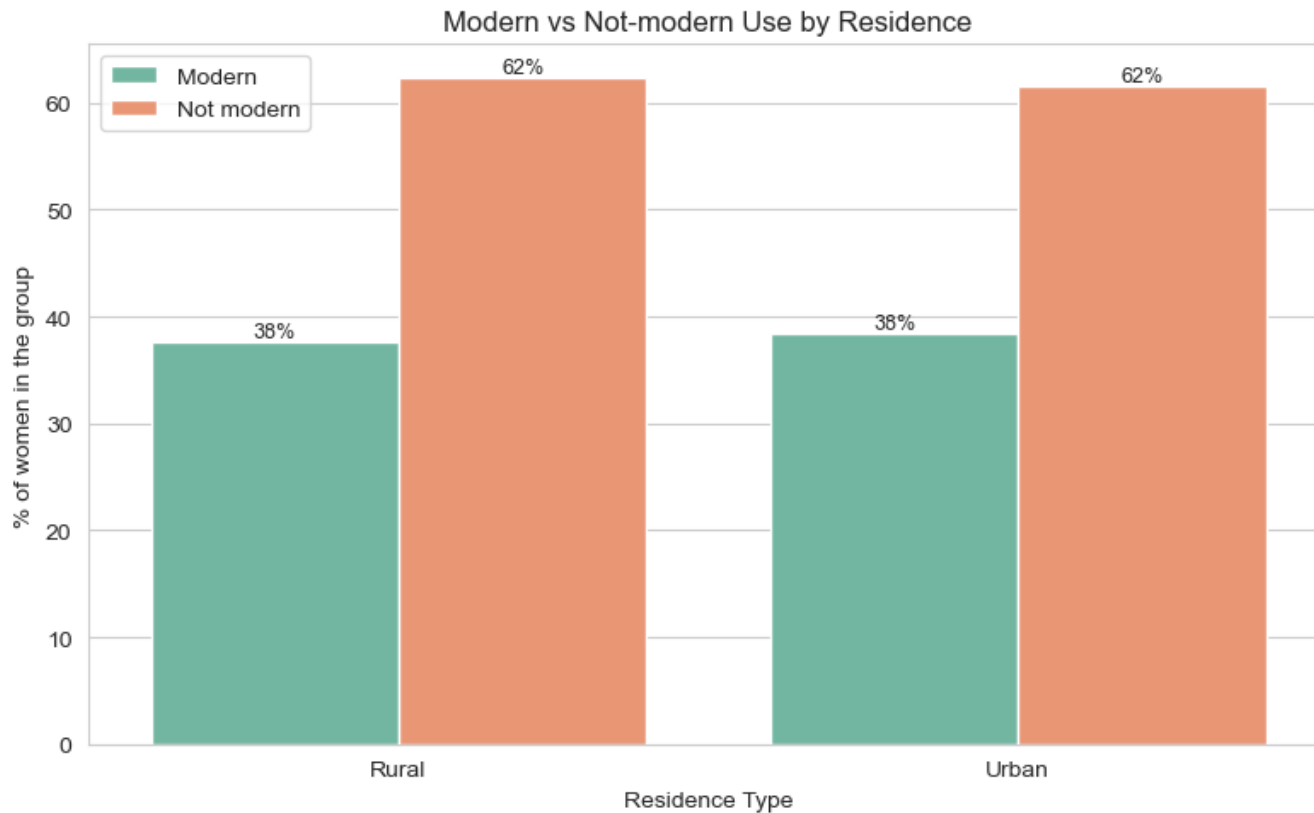
WHAT YOU'RE SEEING

Each cell is the modern-use rate for one wealth × education group. With no education, use stays at 9–21% no matter how wealthy the household. With any schooling, it jumps into the 29–48% range across every wealth group.

WHY IT MATTERS

The full answer to Q3: education is the stronger lever. Raising incomes alone would probably not be enough — the no-education row stays pale all the way across. Programmes that keep girls in school, and adult literacy for women, are also family planning programmes.

Town vs country: a myth, tested



WHAT YOU'RE SEEING

Urban and rural women use modern methods at almost exactly the same rate — about 38% each.

WHY IT MATTERS

Q4 answered: on its own, where a woman lives barely matters. Any urban–rural gap people expect is explained by education and wealth, not by residence itself. That is genuinely useful for planners: don't budget by town-versus-country — budget by county, poverty, and education, where the real differences live.

Four findings that shaped what we built

Two rare groups

Found: Traditional (3.8%) and Folkloric (0.2%, 53 women) are tiny.

So we: Balanced the training data and used metrics that respect small groups — raw accuracy alone would hide failure.

County is the biggest signal

Found: A 56-point spread, with every arid county at the bottom.

So we: Gave the model an explicit arid-county flag so geography is seen directly, not guessed.

Education beats wealth

Found: No-education women stay at 9–21% use at every wealth level.

So we: Kept education and wealth as ordered scales so the cliff and the jump are learnable.

Twin variables

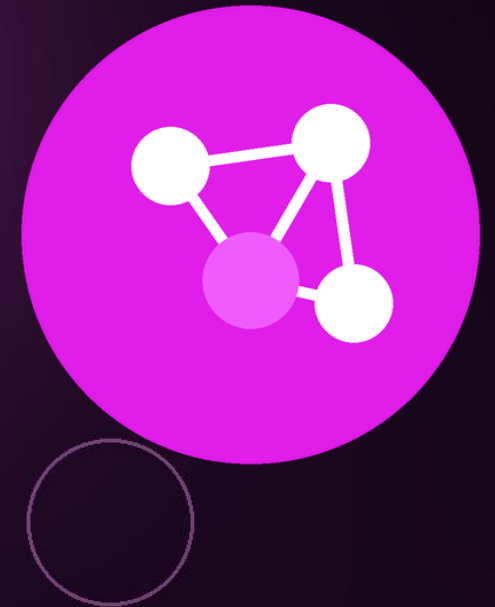
Found: Children ever born $\approx$ living children ($r = 0.98$).

So we: Kept one and derived new features from the difference — no double counting.

PART 3 OF 5

Building the Model

Five different approaches — from a simple statistical baseline to two deep-learning designs — compete on the same held-out test of 6,432 women.



Playing fair with the data



Decode everything

Raw survey codes became words (county 47 → “Nairobi”), so every chart and explanation downstream is readable by the health planners who will use it.



Respect the blanks

Missing ages-at-first-birth belong to women who never gave birth — that's information, not noise. It became a `has_given_birth` flag; partner details for single women became “No partner”, not a guess.



Balance only the practice data

The rare groups were boosted (SMOTE) in training data only — the test set stays exactly as real life looks, so scores are honest.



Engineer what the EDA taught

New features came straight from the findings: an arid-county flag, ordered wealth and education scales, and family-stage measures like child density — each one traceable to a chart in Part 2.

Five models, one held-out test

MODEL	ACCURACY	MACRO F1	MACRO AUC
Logistic Regression (baseline)	54.8%	0.354	0.764
Random Forest	70.6%	0.390	0.775
LightGBM (tuned) — selected	72.4%	0.367	0.798
Deep MLP (neural network)	70.3%	0.385	0.751
TabTransformer	64.8%	0.384	0.754

Runner-up strengths: Random Forest has the best macro F1 (0.390); TabTransformer alone finds 34% of Traditional users.

WHY LIGHTGBM WON

Best accuracy (72.4%) and the best ability to tell the four groups apart (macro AUC 0.798) — plus practical wins that matter for a Ministry of Health deployment:

- Explains itself — native SHAP support
- Tiny (6 MB) and fast — predictions in under a millisecond
- No special hardware or deep-learning software needed

THE HONEST TRADE-OFF

On the two big groups — 96% of all women — the models are strong. On the two rare groups (Traditional, Folkloric) every one of the five architectures hits the same wall. That wall is the data, not the models — and it shaped our recommendations.

Tuning the winner: searching smarter, not harder

How the search worked

Instead of trying every combination (nearly two million), Optuna uses Bayesian search — like asking locals for restaurant tips after each meal instead of wandering the city at random.

30 trials · quick 2-fold checks · a fast 15,000-row subsample — then the winning recipe is retrained on all 59,820 training rows.

The winning recipe

Trees

700

Learning rate

0.011

Max depth

10

Leaves per tree

70

The payoff — before vs after tuning (held-out test set)

ACCURACY

+23.5 pp

72.4% → **95.9%**

MACRO F1

+0.143

0.367 → **0.510**

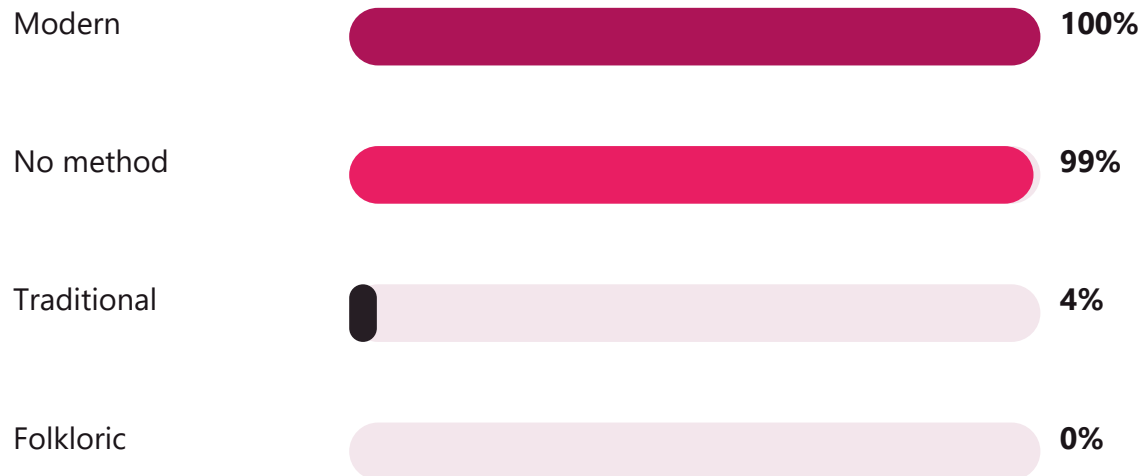
MACRO AUC

+0.153

0.798 → **0.951**

The final tuned model, up close

How often it finds each group (recall, test set)



Strong on the 96% of women in the two main groups; honest about the rest.

THE HEADLINE SCORES

95.9% accuracy · 0.951 macro AUC · 0.510 macro F1 on 6,432 women the model never saw during training — the tuned configuration (700 trees, learning rate 0.011) carried into the deployed pipeline.

USE IT AS A RISK FLAG

Stakeholders should read its output as a two-way flag — “likely non-user” vs “likely modern user”. Pretending four-class precision would create false confidence exactly where the data is weakest.

PART 4 OF 5

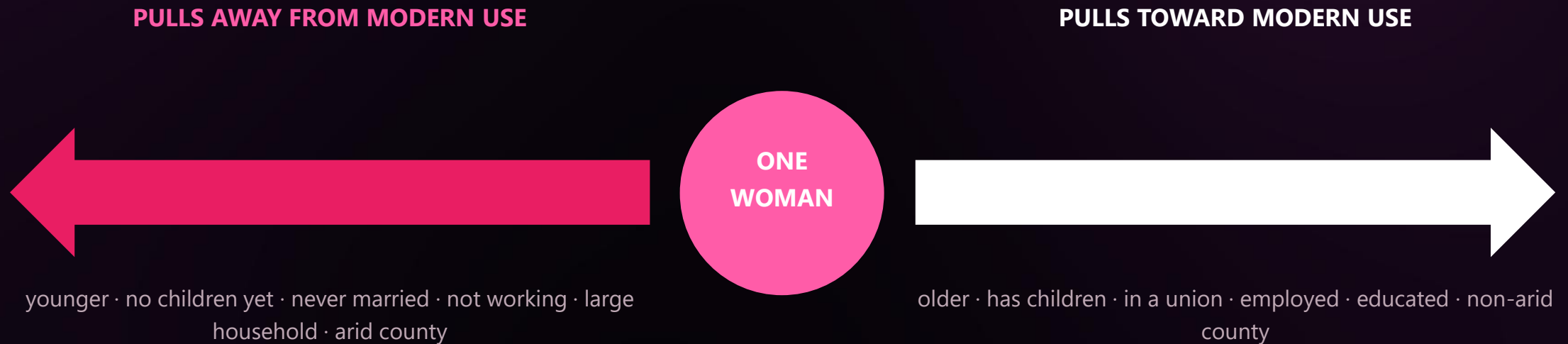
Why the Model Decides

A prediction nobody can question is a prediction nobody should trust. SHAP opens the box — no statistics degree needed.



Think of it as a tug-of-war

For every single woman, each fact about her life pulls the prediction one way or the other. SHAP measures exactly how hard each fact pulls.



Read it like a before-and-after comparison — no statistics background required.

What actually drives the prediction

1

Age — the strongest driver

A life-stage effect: older women are pushed strongly toward modern use; younger women away. Not something policy can change — but essential for timing outreach.

2

Living children — a close second

Women adopt contraception after having children, not before — exactly what the EDA showed. Postpartum contact is the moment advice lands.

3

Employment — the actionable lever

Employed women are pushed toward modern use. Unlike age or children, employment can be changed by programmes — the most policy-tractable finding in the project.

4

Arid county — a genuine barrier

Living in an ASAL county pushes away from modern use even after wealth, education and religion are accounted for. Real access and infrastructure gaps, not a proxy for poverty.

5

Being in a union — the top enabler

The single strongest enabler of modern use in the per-class analysis — and, paired with the other factors, the key to the doorstep checklist in Part 5.

6

Household size — the surprise

Larger households push away from modern use — suggesting stretched resources rather than greater demand.

Two real women — how the model decided

RESPONDENT C

Predicted: “No method” — 99.9% sure

Young · never married · no children · not employed

Every strong pull comes from her life stage: she is not yet exposed to pregnancy risk at all.

The model is not flagging an access problem — it is recognising a woman who does not need an intervention visit. Sending outreach to her doorstep would waste everyone's time.

RESPONDENT D

Predicted: “Modern” — 92.5% sure

Older · married · employed · several children · non-arid county

Every lever points the same way — union, employment, family stage, geography.

She is the prototype of the woman today's programmes already reach. The women our recommendations target are the ones missing these enablers — exposed to risk, but without the supports Respondent D has.

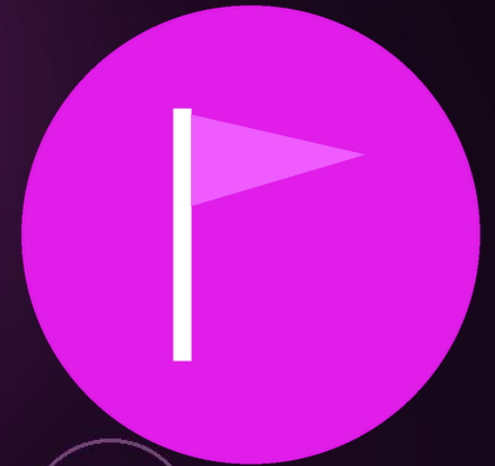
THE POINT

Telling these two apart from a high-risk non-user is exactly what the model is for — target the women who are exposed to risk but missing the enablers.

PART 5 OF 5

From Insight to Action

Analysis that stays in a notebook helps no one. Here is what we now know, what we recommend, and the tool that is already live.



Six things we now know

1

County is the biggest signal in the data

Modern use runs from ~1% in Mandera to 57% in Embu — the arid north needs its own approach.

2

No education means being left behind

Roughly 12% modern use without schooling vs 37–46% with any — and wealth can't close that gap alone.

3

Wealth acts like a cliff

25% for the poorest fifth, then flat at 41–43% for everyone else.

4

Where you live barely matters — who you are does

Urban and rural women both sit at ~38%; county, poverty and schooling drive the real differences.

5

The best model improved with tuning — and stays honest

LightGBM won the five-model contest and Optuna tuning lifted it to 95.9% accuracy / 0.951 macro AUC — used as a two-way risk flag, not a four-class oracle.

6

The drivers are knowable and usable

Age and children lead; union status enables; arid geography blocks; employment is the lever programmes can pull.

What we recommend — and to whom

Give the arid north its own programme tier

MOH · UNFPA

Mandera, Wajir, Garissa, Marsabit and Tana River face real infrastructure barriers — dedicated budget and staffing, not a share of a uniform plan.

Use the four-question doorstep checklist

COMMUNITY HEALTH PROMOTERS

In a union + not employed + poorest wealth group + arid county = priority visit. No model, smartphone or training required.

Fund women's employment as family planning

COUNTIES · NGOS

Employment is the strongest driver a programme can actually change — co-locate family planning with microfinance and vocational training.

Aim subsidies at the poorest fifth

MOH · USAID

The wealth cliff means targeted free supplies beat universal subsidies on impact per shilling.

Don't chase the rare classes

ALL PARTNERS

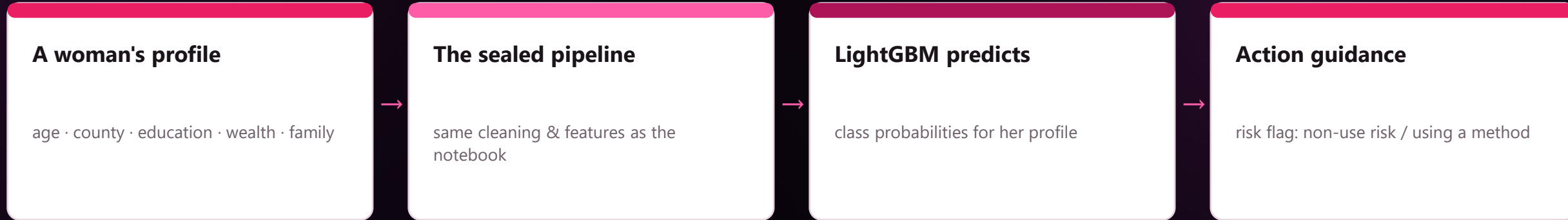
Treat the model as a two-way risk flag. Promising four-class precision would fail exactly where the data is weakest — and cost trust.

Put the live tool in the workflow

CHMTS · FP2030

The API returns a prediction and risk flag for any profile — usable from a browser today, or integrated into county dashboards (DHIS2).

It's not a slide — it's already live, and it works



VERIFIED END-TO-END — A REAL TEST CASE

We sent a real profile through the live system: a 28-year-old married woman from Turkana — an arid county — primary-educated, in the poorest wealth group, unemployed, with three children.

The model flagged her as a non-use risk (52% “No method” vs 41% “Modern”) — exactly the profile the analysis says outreach should prioritise. The prediction, the reasoning and the recommendation all point the same way.

kdhs-contraceptive-api.onrender.com

THE INSIGHT ARCHITECTS

Thank you

Questions & discussion welcome.

- Claire Simiyu
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- Joy Nyuguto
- Antony Sila
- Shadrack Symekah
- Martin Kitema

Live tool: kdhs-contraceptive-api.onrender.com

Code & data: github.com/symekah1999 — KDHS 2022 Capstone

